

ADDENDUM FOR THE PAPER “EPSTEIN-POINCARÉ SURFACES FOR G -OPERS”

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1. INTRODUCTION

The purpose of this note is to remark that one can cheaply improve the results in my paper by either slightly modifying some proofs or ignoring a claim I made. My hope is that this will be incorporated into the main paper. I am temporarily publishing this on my website to avoid clashing with the publication process and, at the same time, to show whoever reads my research statement that I am not bluffing. We will import all the notations from the main paper [Lem26].

The following supersedes Theorem 1.9 and answers Question 1.10 positively. For context, given $x, y \in \mathcal{F}$, two elements in the full flag manifold, we say that they are *transverse* (denoted $x \pitchfork y$) if their G -orbit in $\mathcal{F} \times \mathcal{F}$ is open. Given ρ a quasi-Hitchin representation, we define Ω^{thin} to be the complement of the *naive thickening* of the limit set Λ_ρ , defined as $\text{Th}^{naive}(\Lambda_\rho) = \bigcup_{\xi \in \Lambda_\rho} \mathcal{Z}_\xi$, for $\mathcal{Z}_\xi$ is the set of all points non-transverse to ξ (i.e., the Schubert loci).

Theorem 1.1. *Let $(\mathcal{D}, \rho)$ be a quasi-Hitchin G -oper over a compact hyperbolic Riemann surface X , then:*

- (1) *The image of the developing map is contained in Ω^{thin} . Particularly, it is contained in any cocompact domain of discontinuity.*
- (2) *$\mathcal{D}$ extends to a continuous map $\overline{\mathcal{D}} : \tilde{X} \cup \partial\tilde{X} \rightarrow \mathcal{F}$, by defining $\overline{\mathcal{D}}|_{\partial\tilde{X}} = \xi$, the boundary map of the quasi-Hitchin representation.*

We can use this result to prove two interesting consequences. The following gives a positive answer to Question 1.15.

Theorem 1.2. *Let $(\mathcal{D}, \rho)$ be a quasi-Hitchin G -oper over a Riemann surface X , then $\mathcal{D}$ is transversal in the sense that for any $z \neq w$ we get that $\mathcal{D}(z) \pitchfork \mathcal{D}(w)$. In particular, the developing map is a holomorphic embedding.*

Lastly, Filip’s work [Fil21] made me aware that one can use the computation introduced in [EKMZ18] to compute Lyapunov exponents of flat holomorphic bundles, whenever we get some holomorphic reduction of structure. Costantini [Cos20] further explored this, and applying his result leads to the following.

To any oper $(\mathcal{D}, \rho)$, one can associate the flat G -bundle $\tilde{X} \times G / \sim =: P_G \rightarrow X$. Choosing ω a fundamental weight (and maybe taking a double cover of G) allows us to consider a flat vector bundle $P_G[V_\omega] \rightarrow X$, where V_ω is the G -representation with highest weight ω . We will denote the top Lyapunov exponent of this bundle by $\lambda^\omega(\rho)$.

Theorem 1.3. *Let $(\mathcal{D}, \rho)$ be a G -oper, then for any fundamental weight ω , we get that:*

- (1) *$\lambda^\omega(\rho) \geq \lambda^\omega(\rho_F)$, for ρ_F the holonomy of the Fuchsian G -oper over X . With the curvature normalization of the paper (equal to -2), $\lambda^\omega(\rho_F) = \sqrt{2}\omega(h)$ for h the semisimple element of a Kostant triplet.*

(2) *The equality is attained whenever $(\mathcal{D}, \rho)$ is quasi-Hitchin.*

The first item is a direct application of [EKMZ18], and the second has some merit. Together with results by Daniel-Deroin [DD19], this implies some interesting consequences about the complex geometry of the slice of quasi-Hitchin opers, which is worth exploring in upcoming work.

Finally, an embarrassing omission on my end. Right after Theorem 1.5, I claimed that the neighborhood of quasi-Hitchin opers we obtain around the Fuchsian locus is not uniform, i.e., that it depends on the base Riemann surface. This is wrong, and it follows from a personal confusion in the Cauchy estimates section. More precisely, in Corollary B.4, I obtained the bound $\|\nabla\alpha\|_\infty \leq C(X) \sup_X \|\alpha\|$, for C a constant depending on X . However, when we look at the proof, we observe that one does not need to take into account the injectivity radius, and one in fact obtains the bound

$$\|\nabla\alpha\|_\infty \leq \frac{1}{2}\sqrt{2l+1}\left(1 + \frac{1}{2l}\right)\|\alpha\|,$$

everywhere. The injectivity radius would have played a role if we were considering L^2 norms, but luckily we are not.

This observation answers Question 1.8 affirmatively.

2. PROOF OF THEOREM 1.1

We will use the fact that if $(\rho_n) \in \text{Hom}(\Gamma, G)^\mathbb{N}$ is a sequence of Δ -Anosov representations converging to a Δ -Anosov representation ρ , then the limit maps $\xi_n : \partial\tilde{X} \rightarrow \mathcal{F}$ converge to the limit map of ρ .

Theorem 2.1. *Let $(\mathcal{D}, \rho)$ be a quasi-Hitchin G -oper over a compact hyperbolic Riemann surface X , then:*

- (1) *The image of the developing map is contained in Ω^{thin} . Particularly, it is contained in any cocompact domain of discontinuity.*
- (2) *$\mathcal{D}$ extends to a continuous map $\overline{\mathcal{D}} : \tilde{X} \cup \partial\tilde{X} \rightarrow \mathcal{F}$, by defining $\overline{\mathcal{D}}|_{\partial\tilde{X}} = \xi$, the boundary map of the quasi-Hitchin representation.*

Proof. (1) Notice that, by design, the space of quasi-Hitchin G -opers is connected. We will prove that the set:

$$E = \{(\mathcal{D}, \rho) \in \widetilde{\mathcal{QH}}(X) : \mathcal{D}(\tilde{X}) \subset \mathcal{F} \setminus \text{Th}^{\text{naive}}(\Lambda_\rho)\},$$

is open and closed in the set of quasi-Hitchin opers. By Proposition 8.4, it is non-empty.

We start by verifying openness. Choose $F \subset \tilde{X}$ a fundamental domain for the deck action. If $(\mathcal{D}, \rho) \in E$, we get that $\mathcal{D}(F)$ does not intersect the compact set $\text{Th}^{\text{naive}}(\Lambda_\rho)$, thus it is a finite distance away from it (with respect to some choice of Riemannian metric). As the limit set varies continuously with the representations, and the thickening operation is continuous with respect to the Hausdorff topology (see [KLP17, Lemma 7.3]) we get that any oper $(\mathcal{D}', \rho')$ near D will verify that $\mathcal{D}'(F)$ is disjoint from $\text{Th}^{\text{naive}}(\Lambda_{\rho'})$, thus $\mathcal{D}'(\tilde{X}) \subset \mathcal{F} \setminus \text{Th}^{\text{naive}}(\Lambda_{\rho'})$ as desired by $\rho'(\pi)$ invariance.

We now show that E is closed. Let $(\mathcal{D}_n, \rho_n) \in \widetilde{\mathcal{QH}}(X)^\mathbb{N}$ be quasi-Hitchin opers such that $(\mathcal{D}_n, \rho_n) \in E$ for all n , and $(\mathcal{D}_n, \rho_n) \rightarrow_n (\mathcal{D}, \rho) \in \widetilde{\mathcal{QH}}(X)$. Given $x \in \partial\tilde{X}$ and ξ_n the limit map of ρ_n , define $\sigma_{x,n}^i : G \rightarrow \mathbb{C}$ the holomorphic function defining the section in $\mathcal{F}$ vanishing at $\mathcal{Z}_{\xi_n(x)}^i \in \mathcal{F}$ for $i = 1, \dots, l$ (see Section 8.1). Lifting the oper maps to $\overline{\mathcal{D}}_n : \tilde{X} \rightarrow G$ using the Drinfeld-Sokolov normal form (see Section 3.3), we get that $\overline{\mathcal{D}}_n$ converges (uniformly on compact sets) to the lift $\overline{\mathcal{D}}$ of $\mathcal{D}$.

Given $i \in \{1, \dots, l\}$, the functions:

$$F_n^i(z, x) = \sigma_{x,n}^i \circ \overline{D}_n(z),$$

are holomorphic in $z \in \tilde{X}$, and continuous in x . By assumption, $F_n^i(z, x)$ is non-vanishing for all $n \in \mathbb{N}$. To conclude the proof, recall the classical result of Hurwitz: a uniform limit of a non-vanishing sequence of holomorphic functions is either non-vanishing or identically zero (by the argument principle). Applying this to $F_n^i(z, x)$, we obtain that $\mathcal{D}$ is either contained in $\mathcal{Z}_x$, or $D \in E$. The first case cannot happen, as no smooth point of $\mathcal{Z}_x$ can be tangent to the set of principal directions (see Remark 8.3).

(2) It suffices to see that given $(z_n) \in \tilde{X}^{\mathbb{N}}$ such that $z_n \rightarrow x \in \partial\tilde{X}$, we get that $\mathcal{D}(z_n) \rightarrow \xi(x)$. Pick F a fundamental domain in $\tilde{X}$, and let $(\gamma_n) \in \pi_1(X)^{\mathbb{N}}$ elements such that $z_n \in \gamma_n F$. Observe that $\gamma_n \rightarrow x \in \partial\pi_1(X)$, and without loss of generality, we can assume that $\gamma_n^{-1} \rightarrow y \in \partial\pi_1(X)$. It was proved in [KLP17] that up to taking subsequences, we can assume that the maps $\rho(\gamma_n)$ acting on $\mathcal{F}$ converge uniformly on compact sets of $\mathcal{F} \setminus \mathcal{Z}_{\xi(y)}$ to the constant map to $\xi(x)$. By the previous part, $\mathcal{D}(F)$ is compact and disjoint from $\mathcal{Z}_{\xi(y)}$. In particular, $\rho(\gamma_n)\mathcal{D}(F) \rightarrow \{\xi(x)\}$ in the Hausdorff topology by the uniform convergence, which proves that $\mathcal{D}(z_n) \rightarrow \xi(x)$. $\square$

3. PROOF OF THE CONSEQUENCES

3.1. Transversality of quasi-Hitchin opers.

Theorem 3.1. *Let $(\mathcal{D}, \rho)$ be a quasi-Hitchin G -oper over a Riemann surface X , then $\mathcal{D}$ is transversal in the sense that for any $z \neq w$ we get that $\mathcal{D}(z) \pitchfork \mathcal{D}(w)$. In particular, the developing map is a holomorphic embedding.*

Proof. Let $T \subseteq \widetilde{\mathcal{QH}}(X)$ be the set of transverse quasi-Hitchin opers. This set is non-empty, as the Fuchsian locus is in T . We will prove that T is open and closed following a scheme similar to Theorem 1.1.

We start by showing that T is closed. To see it, consider $(\mathcal{D}_n, \rho_n) \in T^{\mathbb{N}}$ converging to a quasi-Hitchin oper $(\mathcal{D}, \rho)$ and consider $F_n^i(z, w) = \sigma_{\mathcal{D}_n(z)}^i \circ \overline{D}_n(w)$, where σ_x^i denotes the lift to G of the section of $\mathcal{F}$ with zeros in $\mathcal{Z}_x^i$ for $i = 1, \dots, l$ as before. The functions $F_n^i : \tilde{X} \times \tilde{X} \setminus \Delta \rightarrow \mathbb{C}$ are holomorphic, where Δ is the diagonal, and non-vanishing for all n by assumption. Moreover, they converge to $F^i(z, w) = \sigma_{\mathcal{D}(z)}^i \circ \overline{D}(w)$. Hurwitz’s theorem remains valid in several variables (by applying it to each slice). Thus, F^i is either non-vanishing or identically zero. However, the second case cannot occur because opers are locally transverse (Theorem 1.13), so $\mathcal{D}$ is transverse.

We now show that $\widetilde{\mathcal{QH}}(X) \setminus T$ is closed as well. Consider $(\mathcal{D}_n, \rho_n)$ in the complement of T converging to $(\mathcal{D}, \rho) \in \widetilde{\mathcal{QH}}(X)$. Defining functions $F_n^i(z, w)$ and F^i as in the previous paragraph, we get that there must be $(z_n, w_n) \in \tilde{X} \times \tilde{X} \setminus \Delta$ and i for which $F_n^i(z_n, w_n) = 0$ for all n . Moreover, we can assume that z_n lies in a fundamental domain F for the $\pi_1(X)$ action. We now have several possibilities:

- $\exists C > 0$ so that $\frac{1}{C} < d(z_n, w_n) < C$. In this case, up to taking subsequences, $w_n \rightarrow w$, and $F^i(z, w) = 0$; thus, the limit $\mathcal{D}$ is not transverse.
- $d(z_n, w_n) \rightarrow_n 0$. This cannot happen because $\mathcal{D}$ is locally transversal. Whenever a sequence of holomorphic functions converges uniformly on a compact set to a limit, then its derivatives also converge. This implies that there is some $\varepsilon > 0$ around $\lim z_n = z$, so that $\mathcal{D}_n|_{B(z_n, \varepsilon)}$ is locally transverse for all n , which contradicts the fact that $d(z_n, w_n) \rightarrow 0$.

- Finally, we get the situation when $d(z_n, w_n) \rightarrow \infty$. Without loss of generality, $w_n \rightarrow x \in \partial\tilde{X}$, and $z_n \rightarrow z$. If this is the case, we get that $\mathcal{D}_n(z_n) \in \mathcal{Z}_{\mathcal{D}_n(w_n)}$ for all n , thus $\mathcal{D}(z) \in \mathcal{Z}_{\xi(x)}$, for ξ the limit map of ρ by because $\mathcal{D}$ can be extended continuously to the boundary. This again contradicts the fact that $\mathcal{D}(\tilde{X})$ does not intersect the naive thickening of the limit set.

□

The proof above shows that the closedness argument does not use the quasi-Hitchin hypothesis.

3.2. Lyapunov exponents. Given a compact hyperbolic Riemann surface X with curvature -2 , we denote by UX its unit tangent bundle, by $g_t : UX \rightarrow UX$ the geodesic flow, and by μ its Liouville measure on UX .

Let (E, ∇) be a flat complex vector bundle over X , and let $\bar{E} \rightarrow UX$ be its pullback bundle. Parallel transport along ∇ lifts g_t to a flow $G_t : \bar{E} \rightarrow \bar{E}$. Fix any continuous Hermitian metric $\|\cdot\|$ on E . Oseledets' theorem (see for instance [Fil19]) provides constants $\lambda_1(E, \nabla) \geq \dots \geq \lambda_{\dim E}(E, \nabla)$, called the *Lyapunov exponents* of (E, ∇) , and for μ a.e. $x \in UX$ a measurable splitting $E = \bigoplus_{i=1}^{\dim E} E_{\lambda_i(E)}$ for which:

$$\lambda_i(E, \nabla) = \lim_{T \rightarrow \pm\infty} \frac{1}{T} \log \|G_T(v)\|,$$

for any $v \in E_{\lambda_i} \setminus \{0\}$. These exponents do not depend on the choice of the metric.

The following result by Costantini (proven in [DD19] in a slightly different setup) allows us to compute the Lyapunov exponent when E is a holomorphic flat bundle.

Theorem 3.2 ([Cos20, Theorem 3.4, Proposition 3.9]). *Let (E, ∇) be a holomorphic flat bundle over a compact, hyperbolic Riemann surface X .*

- (1) *Given $L \subset E$ a holomorphic line subbundle, we get for a.e. $z \in X$, and Lebesgue a.e. $\alpha \in E_z^*$, we get:*

$$\lambda_1(E, \nabla)/\sqrt{2} \geq \frac{\deg(L)}{\deg(K)} + \text{Err}^L(\alpha),$$

where:

$$\text{Err}^L(\alpha) = \pi \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \frac{\#\{s_L^{-1}(\ker \alpha) \cap D_t\}}{\text{area}(D_t)} dt,$$

for $s_L : \tilde{X} \rightarrow \mathbb{P}(E)$ the developing of the triple (E, L, ∇) based at $z \in X$, and D_t a hyperbolic ball of radius t centered on a lift of z .

- (2) *If there is $\mathcal{S} \subset \mathbb{P}(\bar{E}^*)$ a G_t invariant closed subset, for which $s_L(\tilde{X}) \cap \ker \alpha = \emptyset$, $\forall \alpha \in \mathcal{S}$, and there is $\alpha \in \mathcal{S}$ computing the top Lyapunov exponent of E^* , then:*

$$\lambda_1(E, \nabla)/\sqrt{2} = \frac{\deg(L)}{\deg(K)}.$$

We remark that the $\sqrt{2}$ factor comes from the normalization that the hyperbolic metric has curvature -2 .

For G -bundles, one can use fundamental representations to define a Lyapunov exponent.

Definition 3.3. Let $(\mathcal{D}, \rho)$ be a G -oper over X with developing data (P_G, P_B, ∇) , and let ω be a fundamental weight with irreducible representation $\rho_\omega : G \rightarrow GL(V_\omega)$ (passing to a finite cover of G if necessary). The ω -Lyapunov exponent of the oper is

$$\lambda^\omega(\rho) := \lambda_1(P_G[V_\omega], \nabla^\omega),$$

where ∇^ω is the connection induced from ∇ .

Notice that the B -reduction of structure given by the oper defines the holomorphic line bundle $P_B[\mathbb{C}v_\omega] \subset P_G[V_\omega]$, which we will use in order to apply Theorem 3.2.

Lemma 3.4. *Given (P_G, P_B, ∇) developing data for a G -oper over X , and ω a fundamental weight, we get that the line bundle $P_B[\mathbb{C}v_\omega]$ is isomorphic to $K^{\omega(h)}$, for h the semisimple element of a Kostant triplet.*

Proof. Corollary 3.15 shows that $P_G[\mathfrak{g}]$ is isomorphic to $\bigoplus_{i=-m_l}^{m_l} K^i \otimes \underline{\mathfrak{g}}_i$ where each factor coincides with the associated H -bundle $P_H[\mathfrak{g}_i]$. Write $\omega = \sum_{i=1}^l c_i \alpha_i$ for α_i the simple roots. Since $\mathfrak{g}_{\alpha_i} \subseteq \mathfrak{g}_1$ in terms of the h -grading, therefore $\omega(h) = \sum_{i=1}^l c_i$, as $\alpha_i(h) = 1$ for all i . Notice that

$$P_H[\mathbb{C}v_\omega] \cong \bigoplus_{i=1}^l P_H[\mathfrak{g}_{\alpha_i}]^{c_i} \cong K^{\sum_{i=1}^l c_i},$$

which shows the desired claim. $\square$

The key remark is that whenever the oper is quasi-Hitchin, the error terms vanish thanks to Theorem 1.1.

Theorem 3.5. *Let $(\mathcal{D}, \rho)$ be a G -oper, then for any fundamental weight ω , we get that:*

- (1) $\lambda^\omega(\rho) \geq \lambda^\omega(\rho_F)$, for ρ_F the holonomy of the Fuchsian G -oper over X . With our curvature normalizations, $\lambda^\omega(\rho_F) = \sqrt{2}\omega(h)$ for h the semisimple element of a Kostant triplet.
- (2) The equality is attained whenever $(\mathcal{D}, \rho)$ is quasi-Hitchin.

Proof. (1) Let (P_G, P_B, ∇) be developing data for the G -oper, and let $E = P_G[V_\omega]$ be the associated bundle, for $\rho_\omega : G \rightarrow GL(V_\omega)$ a representation with highest weight given by ω (up to maybe taking a finite cover of G), $L = P_B[\mathbb{C}v_\omega] \subset E$ for v_ω the highest weight vector, and ∇^E the induced flat connection. Lemma 3.4 automatically shows that $\frac{\deg(L)}{\deg(K)}$ is a constant equal to $\omega(h)$, independently of the oper.

(2) To prove the equality for quasi-Hitchin opers, we use the second part of Theorem 3.2. Recall that given $\rho : \pi_1(X) \rightarrow G$ a Δ -Anosov representation, one gets that $\rho_\omega \circ \rho : \pi_1(X) \rightarrow GL(V_\omega)$ is a *projective Anosov representation* (see [GW12, Proposition 4.3]). One can show that the limit sets in $\mathbb{P}(V_\omega)$, and $\mathbb{P}(V_\omega^*)$ are parametrized by limit maps ξ_ω, ξ_ω^* from the boundary of $\mathbb{H}^2$ into $\mathbb{P}(V_\omega)$ and $\mathbb{P}(V_\omega^*)$. Such maps coincide with the projection of the limit map $\xi : \partial\mathbb{H}^2 \rightarrow \mathcal{F}$, composed with the embedding of the full flag manifold into each projective space $\mathbb{P}(V_\omega), \mathbb{P}(V_\omega^*)$ induced by ρ_ω and ρ_{ω^*} respectively.

One can think of $\underline{V}_\omega = U\tilde{X} \times V_\omega$ as the pullback of $\bar{E}$ on UX to $U\tilde{X}$, after trivializing it with the flat connection. Recall that one can identify $U\tilde{X}$ with $\partial^{(2)}\mathbb{H}^2 \times \mathbb{R}$ via the Hopf parametrization mapping a vector x to the endpoints of the corresponding geodesic (x_+, x_-) , and a real number t , which is the distance from the closest point projection of a basepoint to the geodesic defined by x . The limit map from the last paragraph gives us a splitting $(\underline{V}_\omega)_x = \xi_\omega(x_+) \oplus \ker \xi_\omega^*(x_-)$. By the definition of an Anosov representation, this splitting is dominated, which implies that for μ -a.e. $x \in U\mathbb{H}^2$, $\xi_\omega^*(x_-)$ is the top Oseledets line on E^* .

Let $\mathcal{S} \subset \mathbb{P}(\bar{E}^*)$ be the image of the section mapping $x \rightarrow \xi_\omega^*(x_-)$. By construction, $\mathcal{S}$ is G_t -invariant. The previous paragraph shows that $\mathcal{S}$ contains a hyperplane computing the Lyapunov exponents of $\bar{E}^*$. We will now see that $s_L(\tilde{X}) \cap \ker \alpha$ vanishes for any $\alpha \in \mathcal{S}$. Let $s_L : \tilde{X} \rightarrow \mathbb{P}(V_\omega)$ the developing map of (E, L, ∇^E) . This developing map equals the composition of $\mathcal{D}$ with the embedding of $\mathcal{F}$ into $\mathbb{P}(V_\omega)$. Moreover, the thickening of the

limit set maps to the union of the hyperplanes $\xi^*(z)$ for $z \in \partial\mathbb{H}^2$. Particularly, since $\mathcal{D}(\tilde{X})$ is disjoint from the naive thickening of the limit set, we get that $s_L(z)$ is transverse to $\ker \alpha = \xi^*(x_+)$ for every $z \in \tilde{X}$, which implies that $\text{Err}^L(\alpha) = 0$, and we get the desired equality. $\square$

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